

Predicting NBA Scoring from Player and Performance Metrics

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Introduction

Research Question

This paper investigates which player characteristics and performance metrics best explain an NBA player's scoring output. Specifically, we wanted to find out how an athlete's physical trait such as age, height, weight affected a player's points per game (**pts**) as well as on-court statistics such as, games played, rebounds, assists, net rating, usage rate, true shooting percentage, and assist percentage? As a result, our mission for this project is to build a regression model that finds these relationships and identifies / validate which predictors carry independent explanatory power.

Background and Data Source

The data come from a kaggle set we found called "NBA Players", which compiles per-season box-score and information about their physical traits scraped from official NBA sources. The dataset contains 12,844 player-season observations from the 1996 to 2023 seasons. Each row represents one player in one season and has 22 variables per row: identifying fields (player name, team, season, college, country, draft information), physical attributes (age, height in cm, weight in kg), and performance measures (games played, points, rebounds, assists, net rating, offensive/defensive rebound percentages, usage percentage, true shooting percentage, and assist percentage). Our response variable is points per game (**pts**), and the candidate predictors are the numeric physical and performance variables.

Method and Justification

Because the response is a continuous quantity and we want to estimate the simultaneous effect of many predictors while holding the others fixed, we use multiple linear regression. This method is appropriate here because it lets us interpret each estimated coefficient as the marginal association between a predictor and scoring, test individual predictors via t-tests, and assess overall model fit via the F-test and R^2 . We begin by examining a scatter-plot matrix to inspect pairwise linear relationships and possible collinearity, then fit a full model and check the standard regression assumptions (linearity, constant variance, and normality of errors) through diagnostic plots. However there were some cases where the assumption were violated and so during those cases we apply a Box-Cox transformation of the response to stabilize variance and improve normality, justifying the transformation empirically through the estimated power parameter rather than assuming linearity holds on the raw scale.

Overview of the Paper

The remainder of the paper proceeds as follows. We first present exploratory analysis via a scatterplot matrix of the response and predictors. We then fit the full multiple linear regression model and report its coefficient estimates, R^2 , and overall F-test. Next, we assess model adequacy through residual diagnostics, identifying any violations of the regression assumptions. Based on these diagnostics, we determine whether a transformation of the response is needed and estimate the appropriate Box-Cox power. Finally, we perform variable selection to arrive at a reduced model, interpret its coefficients against basketball intuition, and conclude by summarizing our findings along with the model’s implications and limitations.

Data Description

The following dataset contains 12,844 NBA player-season observations from the 1996-97 through 2022-23 seasons. Each row represents one player in one season, and Points per game (`pts`) is the response variable. The 12 candidate explanatory variables are age, player height, player weight, games played, rebounds, assists, net rating, offensive rebound percentage, defensive rebound percentage, usage percentage, true shooting percentage, and assist percentage. To further clarify, height is measured in centimeters, weight is measured in kilograms, and the percentage variables are recorded as proportions.

Table 1. Summary statistics for the response and explanatory variables.

Variable	Mean	SD	Min	Max
Points per game	8.21	6.02	0.00	36.10
Age	27.05	4.34	18.00	44.00
Height (cm)	200.56	9.11	160.02	231.14
Weight (kg)	100.26	12.43	60.33	163.29
Games played	51.15	25.08	1.00	85.00
Rebounds	3.56	2.48	0.00	16.30
Assists	1.82	1.80	0.00	11.70
Net rating	-2.23	12.67	-250.00	300.00
Oreb pct	0.05	0.04	0.00	1.00
Dreb pct	0.14	0.06	0.00	1.00
Usage pct	0.18	0.05	0.00	1.00
TS pct	0.51	0.10	0.00	1.50
Assist pct	0.13	0.09	0.00	1.00

The average player scored 8.21 points per game, while the median was just 6.7 points. Additionally, the points per game is right-skewed because most player-seasons have lower scoring averages while a smaller number have a lot higher values. The response ranges from 0 to 36.1 and includes 161 observations with zero points per game. In addition to points, games played also has a wide range, while rebounds and assists are concentrated at lower values and right-skewed as well.

Figure 1 shows the selected distributions and relationships that are most relevant to the regression analysis, and in this case, points per game has clear positive relationships with rebounds, assists, and usage percentage, with correlations of approximately 0.62, 0.66, and 0.64, respectively. The full scatterplot matrix also shows that some explanatory variables are related to one another. In particular, height and weight move together, as height is negatively related to assists and assist

percentage, and the two rebound percentages tend to increase together as well, suggesting possible multicollinearity when the variables are included in the same multiple regression model.

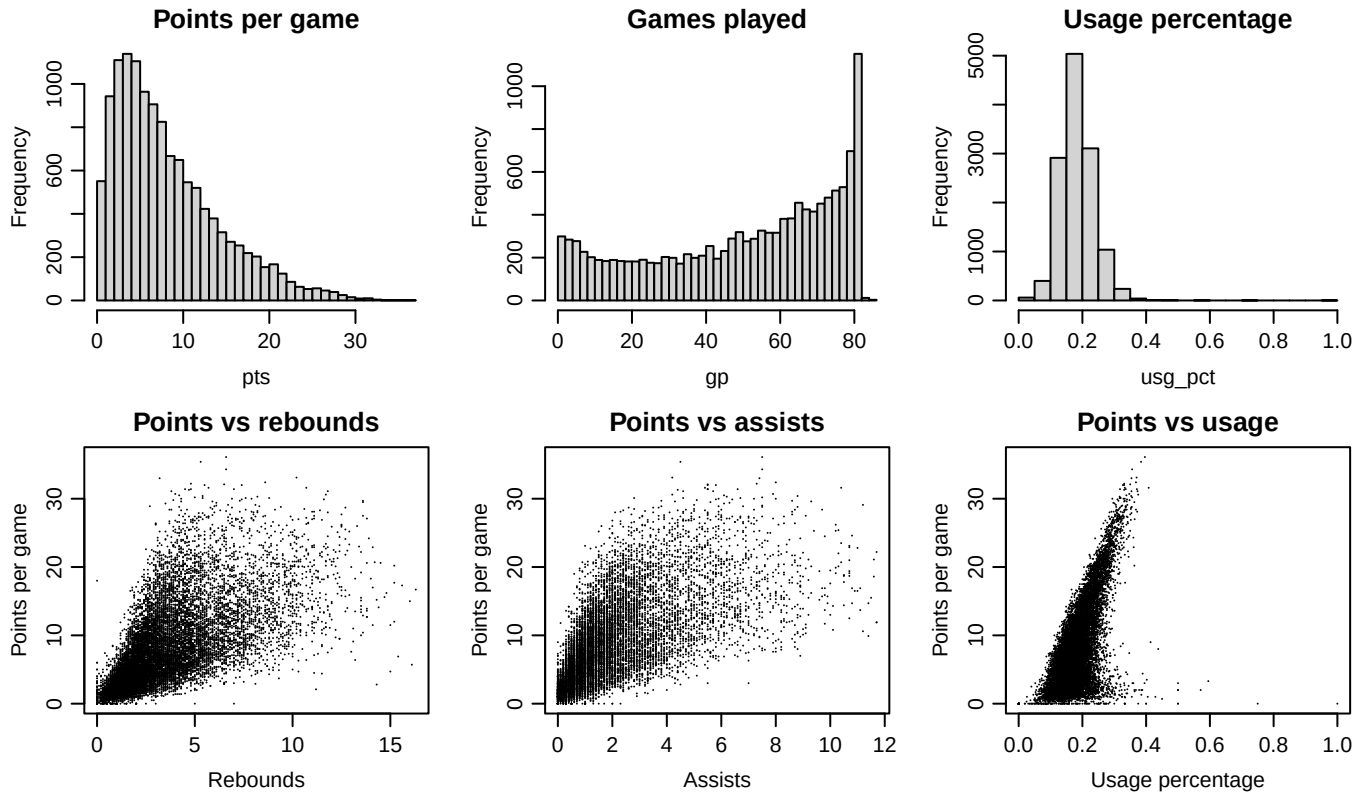


Figure 1: Selected distributions and relationships among the response and explanatory variables.

Results and Interpretation

Candidate Models and Model Selection

To identify the most appropriate predictive model, we evaluated three candidate multiple linear regression models. The first candidate was a full multiple linear regression model that used all twelve explanatory variables (age, player_height, player_weight, gp, reb, ast, net_rating, oreb_pct, dreb_pct, usg_pct, ts_pct, and ast_pct) to predict the response variable, pts. This model served as a baseline because it included all available predictors that were expected to influence a player's scoring. Although the overall model was statistically significant, the diagnostic plots indicated violations of several regression assumptions, including non-constant variance, departures from normality, and evidence of a nonlinear relationship between the fitted values and residuals. These results suggested that the original model could be improved.

To address these issues, we performed variable selection on the untransformed full model using backward elimination. The variables age ($p = 0.60$) and player_weight ($p = 0.98$) were removed because they did not provide significant explanatory power after accounting for the remaining predictors. This produced a reduced set of ten predictors, all of which were significant.

Finally, we addressed the assumption violations by applying a power transformation to the response. Because pts contained zero values, we estimated the transformation parameter using the Box-Cox-with-negatives (bcnPower) procedure, which yielded $\lambda \approx 0.42$; we then applied the corresponding power transformation $\text{pts}^{0.42}$ to the response and refit on the ten retained predictors. The transformation improved the model assumptions by reducing heteroscedasticity and producing residuals that more closely followed a normal distribution, though it did not remove every departure entirely. For these reasons, the reduced transformed model was selected as the best predictive model for the remainder of the analysis.

Final Model: Results and Coefficient Interpretation

After dropping the two predictors that were not significant in the untransformed full model (`age`, $p = 0.60$; `player_weight`, $p = 0.98$) and applying the power transformation $\text{pts}^{0.42}$ to the response, we refit on the remaining ten predictors. Table 2 reports the estimates.

Table 1: Table 2. Final model coefficient estimates (response = $\text{pts}^{0.42}$). All predictors significant at $p < 0.001$.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.7577	0.0717	10.5717	0
player_height	-0.0031	0.0004	-8.7203	0
gp	0.0047	0.0001	45.0381	0
reb	0.1544	0.0015	104.7772	0
ast	0.1593	0.0024	65.7533	0
net_rating	0.0013	0.0002	7.4729	0
oreb_pct	-2.7901	0.0635	-43.9110	0
dreb_pct	-2.1919	0.0507	-43.2168	0
usg_pct	4.8788	0.0427	114.3587	0
ts_pct	1.6790	0.0226	74.2957	0
ast_pct	-1.9630	0.0439	-44.6993	0

The reduced model explains about 91% of the variation in the transformed response ($R^2 = 0.910$, adjusted $R^2 = 0.910$). This is numerically higher than the full untransformed model’s $R^2 = 0.874$, but because the two are measured on different response scales the comparison is only descriptive — the real justification for the transformation is that it improves the normality and non-constant-variance problems identified in the diagnostics, even though some departures remain (see Appendix C). The overall F-test is highly significant ($F = 12,930$ on 10 and 12{,}833 df, $p < 2.2 \times 10^{-16}$), and every retained predictor is significant at any conventional level (all $p < 10^{-13}$), so each contributes independent explanatory power once the others are held fixed.

Because the response is on the $\text{pts}^{0.42}$ scale, each coefficient gives the expected change in *transformed* points for a one-unit increase in that predictor, holding the others constant, rather than a change in raw points per game. The signs and magnitudes still align with basketball intuition: usage rate and true shooting percentage both enter with positive coefficients (both are 0–1 measures, so their per-unit coefficients are large), and rebounds (+0.154) and assists (+0.159) raise scoring as well — players who use more possessions, finish efficiently, and stay involved in the offense score more.

Four coefficients carry counterintuitive negative signs: offensive- and defensive-rebound percentage (-2.79 , -2.19), assist percentage (-1.96), and player height (-0.003). These reversals are consistent with the moderate collinearity flagged in the Data Description: each percentage measure shares information with the raw rebound or assist count already in the model, so once the counts are included the percentages absorb role differences rather than reflecting a true negative effect on scoring. Height's small negative sign fits the same pattern — shorter, higher-usage guards tend to out-score taller players after efficiency and role are controlled for. The variance-inflation factors (Appendix D) are all below 5, indicating that this overlap is moderate rather than a severe multicollinearity problem, so the coefficient estimates remain usable even though these signs should be interpreted with care.

Discussion

Our final regression model revealed the attributes of a player which were the most significant predictors of points per game. In general, it was found that offensive attributes such as usage percentage, assists, and rebounds were the most predictive of a player's scoring ability. Meanwhile, physical attributes contributed little once role and performance were accounted for: player weight was not significant in the full model and was removed during selection, while player height remained statistically significant but had only a very small estimated effect. This demonstrates that a player's role in the offense is more predictive of their scoring ability than their physical attributes. Additionally, our model showed that there were variables in our model which ceased to be significant upon controlling for the other variables. As such, it is essential to perform multiple regression analysis rather than assessing each variable independently.

The results of the study were expected, as attributes such as usage percentage, assists, and rebounds are directly predictive of a player's ability to score. A player with a higher usage percentage will generate more points due to their elevated role in the offensive system, which creates scoring opportunities for themselves and their teammates. Similarly, players who record more assists and rebounds generally stay on the court for longer and are more involved in the game, which increases their scoring opportunities. Meanwhile, physical attributes are less predictive of a player's scoring ability because players of all sizes can be productive scorers depending on their role.

The results of the study have important implications for the sport. They demonstrate that a player's involvement in the offense is the key predictor of their scoring ability. As such, coaches and analysts can utilize these insights to better evaluate players on their scoring potential depending on their role within a team's offensive system.

The main limitation of the study was that it relied on the assumptions of multiple linear regression. While we tested the data for linearity, normality, and homogeneity of variance, a few of these assumptions were not met, and the Box-Cox transformation was applied. Moreover, it is possible that the model missed a few significant variables which have an impact on all players' points per game, such as experience, field goal attempts, three-point percentage, position, and offensive system. These variables should be considered in future research which hopes to build on the findings of this study. Moreover, future research can consider more nuanced models which go beyond linear regression to capture the relationship between the independent and dependent variables.

Appendix

The following plots and numerical measures support the analysis reported in the main text. The complete model summaries, correlation calculations, and Box-Cox output are generated in the accompanying .Rmd file but are omitted from the PDF to avoid repeating results already reported above.

A. Full scatterplot matrix of the response and candidate predictors.

B. Residual diagnostics for the full model. These plots show the departures from normality and the non-constant variance that motivated the transformation.

C. Residual diagnostics for the final model. These plots show that the transformation and predictor removal improved agreement with the regression assumptions.

D. Variance inflation factors for the final model.

Table 2: Table 3. Variance inflation factors for the final model predictors.

	VIF
player_height	2.58
gp	1.64
reb	3.20
ast	4.57
net_rating	1.15
oreb_pct	1.82
dreb_pct	2.41
usg_pct	1.25
ts_pct	1.27
ast_pct	4.11

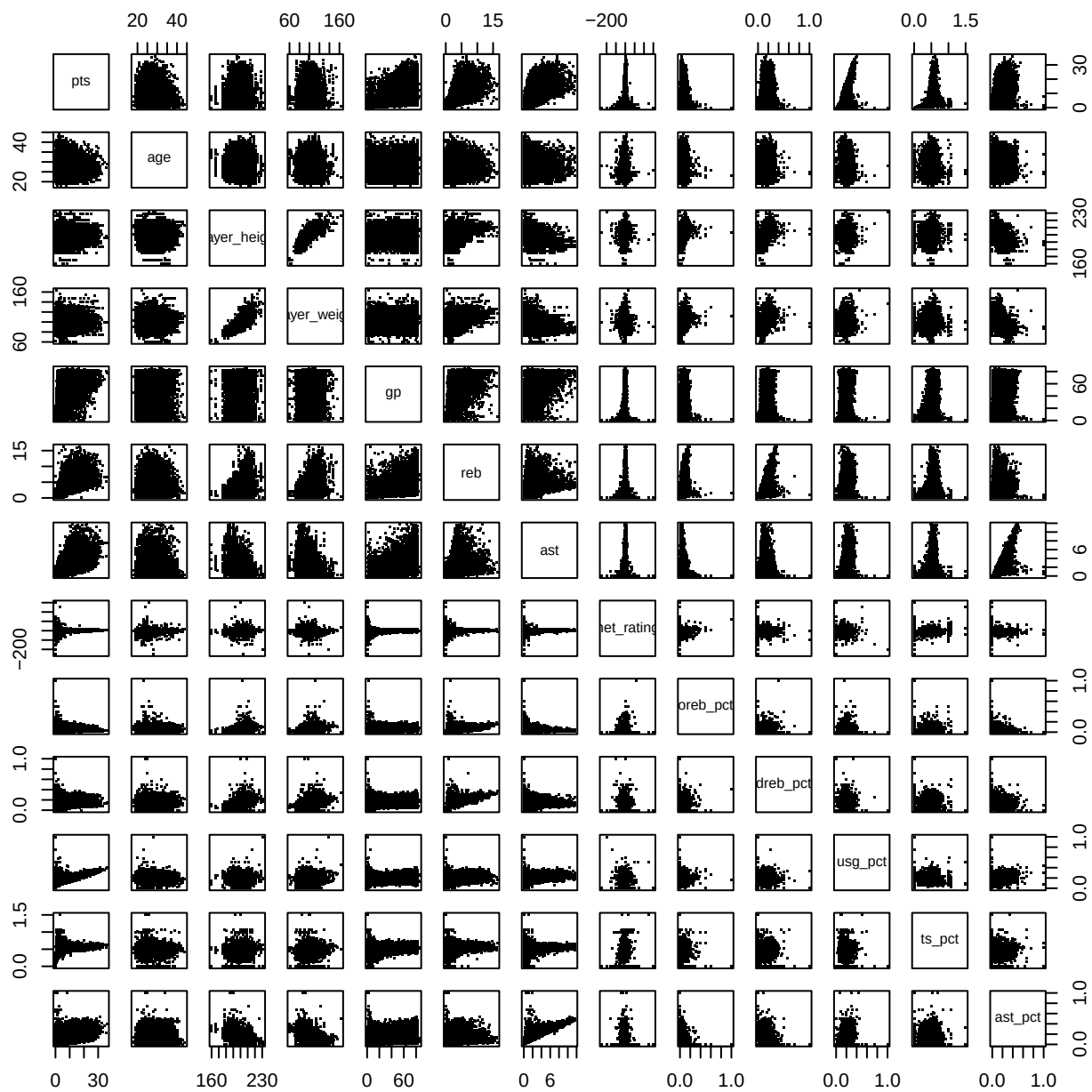


Figure 1: Scatterplot matrix of the response and candidate predictors.

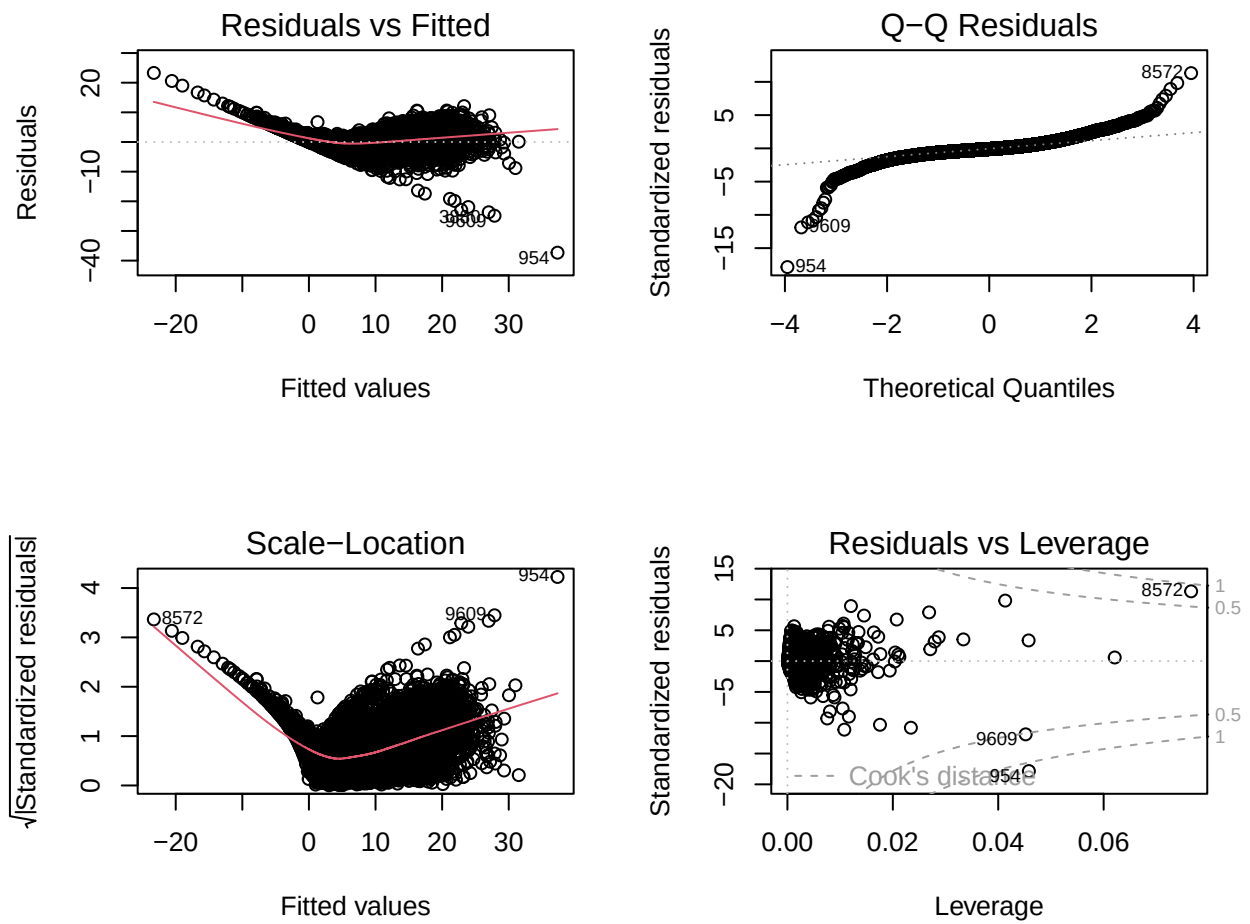


Figure 2: Residual diagnostic plots for the full (untransformed) model.

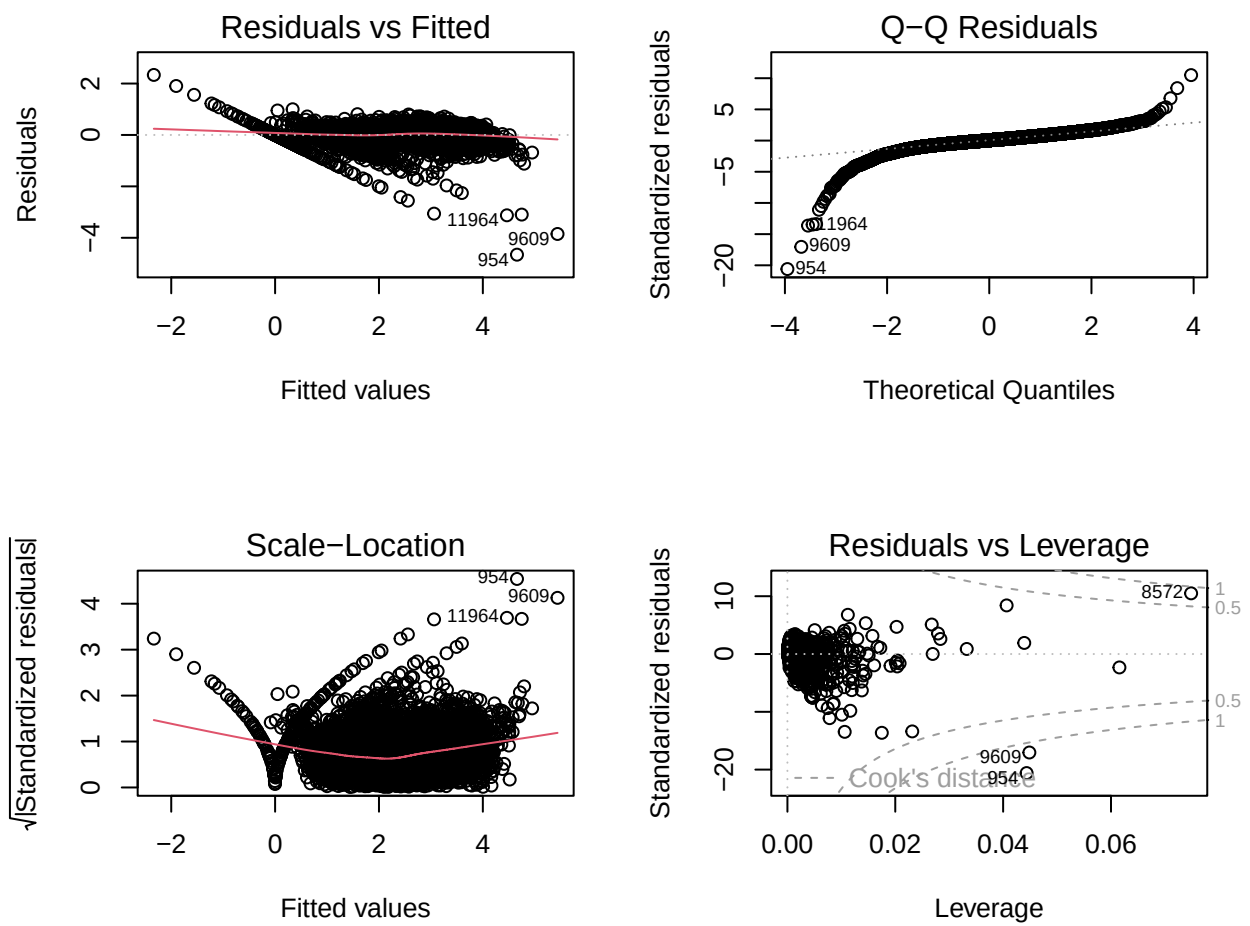


Figure 3: Residual diagnostic plots for the final (transformed) model.